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CSA0503-DATABASE MANAGEMENT SYSTEMS

LAB PRACTICAL

EXP NO:13 Simple programming using CASE and LOOP

Aim:

To learn how to use various MySQL loop statements including case and loop to run a block of code repeatedly based on a condition.

Procedure:

In MySQL, the CASE statement has the functionality of an IF-THEN-ELSE statement and has 2 syntaxes

that we will explore.

CASE Syntax

CASE case_value

WHEN when_value THEN statement_list

[WHEN when_value THEN statement_list] ...

[ELSE statement_list]

END CASE

Procedure:

LOOP Syntax

[begin_label:] LOOP

statement_list

END LOOP [end_label]

LOOP implements a simple loop construct, enabling repeated execution of the statement list, which consists of one or more statements, each terminated by a semicolon (;) statement delimiter. The statements within the loop are repeated until the loop is terminated. Usually, this is accomplished with a LEAVE statement. Within a stored function, RETURN can also be used, which exits the function entirely.

Program 1:

Write a function that uses CASE statement where if monthly_value is equal to or less than 4000, then income_level will be set to 'Low Income'. If monthly_value is equal to or less than

5000, then income_level will be set to 'Avg Income'. Otherwise, income_level will be set to 'High Income'.

Program 2:

Write a function that will use ITERATE statement which would cause the loop to repeat while income is less than 4000. Once income is greater than or equal to 4000, would terminate the LOOP.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE GradeUsingCase(IN marks INT)
-> BEGIN
->   DECLARE grade VARCHAR(10);
->   SET grade = CASE
->     WHEN marks >= 90 THEN 'A'
->     WHEN marks >= 75 THEN 'B'
->     WHEN marks >= 60 THEN 'C'
->     ELSE 'D'
->   END;
->   SELECT CONCAT('Marks: ',marks,' -> Grade: ',grade)
->   AS Result;
-> END //
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
mysql> CALL GradeUsingCase(82);
+-----+
| Result |
+-----+
| Marks: 82 -> Grade: B |
+-----+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE LoopDemo()
-> BEGIN
->   DECLARE i INT DEFAULT 1;
->   simple_loop: LOOP
->     SELECT CONCAT('Loop Count = ',i) AS Looping;
->     SET i = i + 1;
->     IF i > 3 THEN
->       LEAVE simple_loop;
->     END IF;
->   END LOOP;
-> END //
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
mysql> CALL LoopDemo();
+-----+
| Looping |
+-----+
| Loop Count = 1 |
| Loop Count = 2 |
| Loop Count = 3 |
+-----+
3 rows in set (0.00 sec)
```

RESULT:

Thus the Simple programming exercise using CASE and LOOP executed successfully.